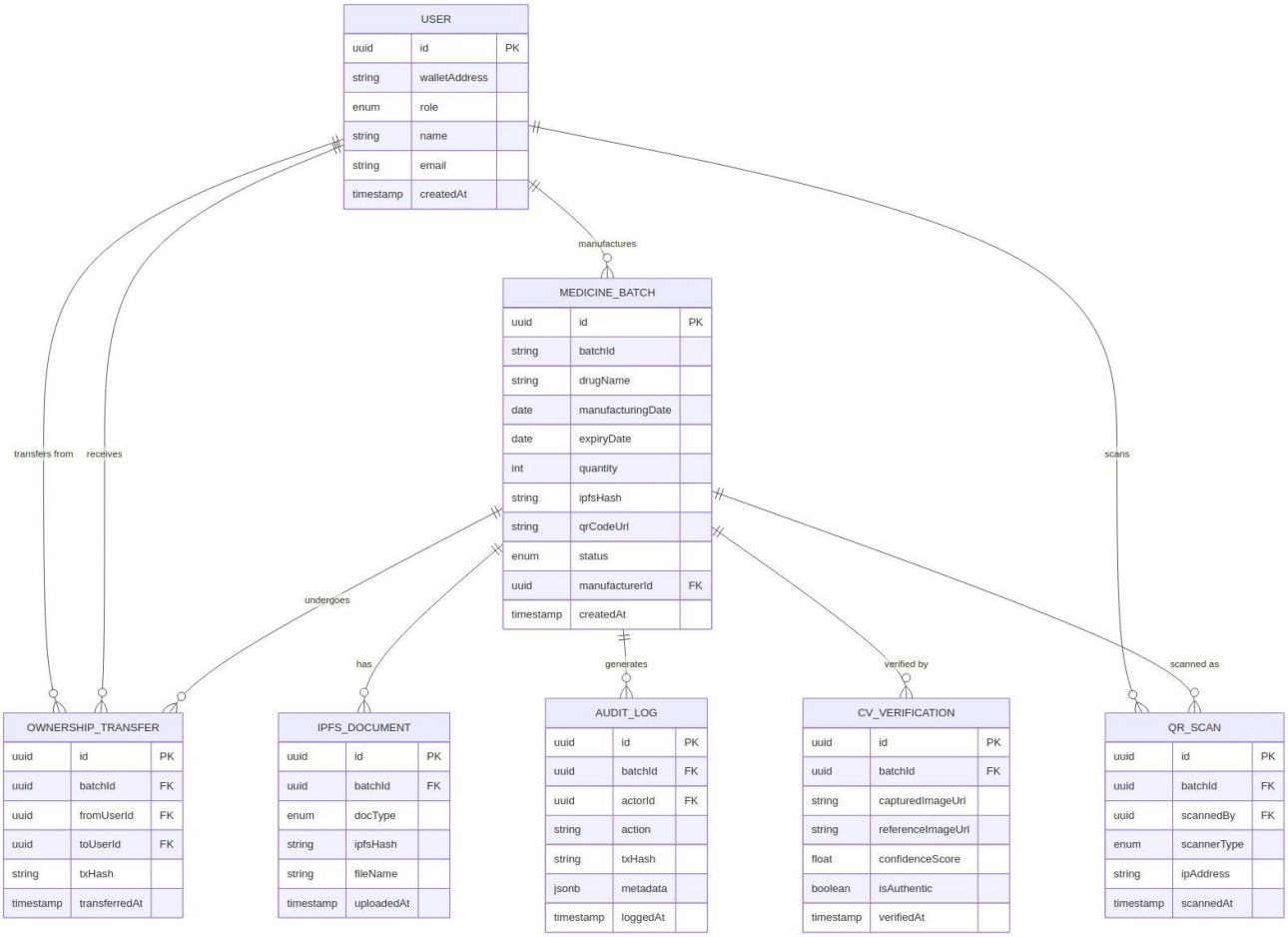
MedSecure – Entity-Relationship (ER) Diagram



Description

The ER Diagram defines the relational database schema underpinning the MedSecure system, specifying all entities, their attributes, primary/foreign key constraints, and inter-entity relationships.

USER: The root entity storing platform participants. Attributes include a UUID primary key (id), walletAddress, role (enum), name, email, and createdAt. A single user may manufacture many batches (one-to-many with MEDICINE_BATCH) and may be involved in multiple scans.

MEDICINE_BATCH: The central entity tracking each drug batch. Attributes include id (PK), batchId, drugName, manufacturingDate, expiryDate, quantity, ipfsHash, qrCodeUrl, status (enum), manufacturerId (FK → USER), and createdAt. A batch undergoes ownership transfers, has IPFS documents, generates audit logs, is verified by CV, and is scanned as QR events.

OWNERSHIP_TRANSFER: Records each custody change. Stores id (PK), batchId (FK), fromUserId (FK), toUserId (FK), txHash, and transferredAt.

IPFS_DOCUMENT: Associates uploaded files with a batch. Stores id (PK), batchId (FK), docType (enum), ipfsHash, fileName, and uploadedAt.

AUDIT_LOG: Captures every significant system event. Attributes include id (PK), batchId (FK), actorId (FK), action, txHash, metadata (jsonb), and loggedAt.

CV_VERIFICATION: Records each computer vision authentication attempt. Stores id (PK), batchId (FK), capturedImageUrl, referenceImageUrl, confidenceScore (float), isAuthentic (boolean), and verifiedAt.

QR_SCAN: Logs each QR scan event. Attributes include id (PK), batchId (FK), scannedBy (FK), scannerType (enum), ipAddress, and scannedAt.